# Chapter 1: Density matrices

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## 1. Basics

- A density matrix represents a broader class of quantum state vector.
- A density matrix can describe isolated parts of systems, such as the state of a system that is entangled with another that we wish to ignore.
- Probabilistic states can be represented by density matrices, allowing both classical and quantum information to be represented by a single mathematical operator.

To say that a $\rho$ matrix is a density matrix, the matrix must have the following properties:

> 1. $Tr(\rho) = 1$, trace of $\rho$.
> 2. The matrix is positive semi-definite, i.e. $\rho \geq 0$. $\rho$ is semi-positive if there exists a matrix M such that:
>
> $$\rho = M^{\dagger} M$$
>
> M can be square or not.

Alternative of semi-positive:

> - $\rho$ is semidefinite positive if $\rho$ is Hermitian, and all its eigenvalues are positive real numbers.
> - $P$ is semidefinite positive if $\langle \psi | \rho | \psi \rangle \geq 0$ for any $|\psi\rangle$ that has the same indices as $\rho$.

Representation of the state matrix of a state of a $|\psi\rangle$:

$$\rho = |\psi\langle\rangle\psi|$$

Here's an example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \begin{pmatrix} \frac{3}{2} & \frac{i}{8} \\ \frac{-i}{8} & \frac{1}{4} \end{pmatrix}$$

Elements on the diagonal represent states in the classical measurement basis, while elements off the diagonal represent the superposition of these states.

Density matrices taking the form $\rho = |\psi\rangle\langle\psi|$ for a quantum state vector $|\psi\rangle$ are known as pure states.Not all density matrices can be written in this form, as some states are not pure.Pure states have a density matrix with $1$ in eigenvalue, and the other eigenvalue are equal to $0$.There is no gloabl phase degeneracy for density matrices, in fact let's suppose:

## 2. Convex combinations

<u>Probabilistic selections</u> :

A convex combination of matrix densities represents a probabilistic selection of quantum states.

If a quantum state $X$ is represented by the state $\rho$ with probability $p$, and by a quantum state $\sigma$ with propability $1 - p$ $(p \in [0, 1])$, the quantum state is represented by a density matrix such that:

$$M = p\rho + (1 - p)\sigma$$

<u>More generally</u>:

$$M = \sum_{k=0}^{m-1} p_k \rho_k$$

This is a convex combination of density matrices $\rho_1, \rho_2, \rho_3, \ldots, \rho_{m-1}$.

For each state this represents :

$$M = \sum_{k=0}^{m-1} p_k |\psi_k\rangle\langle\psi_k|$$

<u>Completely mixed states</u>:

$$\frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{\mathbb{I}}{2}$$

This state represents a completely uncertain state of a qubit.

It does not work for :

$$|\psi\rangle = |\psi_1\rangle + |\psi_2\rangle$$

Indeed, this can be seen by assuming that it's the same vector different from a global phase of $\pi$ and therefore the sum is $0$, whereas this is false.

**Spectral theory:**

Let $M$ be a normal matrix $n \times n$ a complex matrix.

There exists an orthonormal basis of dimension $M$ of complex vectors $\{|\psi_0\rangle, \ldots, |\psi_{n-1}\rangle\}$ with complex numbers $\lambda_0, \lambda_1, \ldots, \lambda_{n-1}$ such that:

$$M = \lambda_0|\psi_0\rangle\langle\psi_0| + \ldots + \lambda_{n-1}|\psi_{n-1}\rangle\langle\psi_{n-1}|$$

decomposition on $\rho = \sum p_k \rho_k$

$$\rho = \lambda_0|\psi_0\rangle\langle\psi_0| + \ldots + \lambda_{n-1}|\psi_{n-1}\rangle\langle\psi_{n-1}|$$

The $\lambda_k$ are the eigenvalues of $\rho$. Since $\rho$ is positive semidefinite

# 3. Bloch sphere

There's a useful geometric method for representing the pure states of qubits, called the *Bloch sphere*. Very practical, it unfortunately only works for qubits: as soon as our system has three or more classical states, the analogous representation no longer corresponds to a spherical object.

At a global pres phase, any $|\psi\rangle$ vector is equivalent to :

$$|\psi\rangle = cos(\theta/2)|0\rangle + e^{i\theta} sin(\theta/2)|1\rangle$$

with $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ .
If $\theta = 0$ or $\theta = \pi$, then $\phi$ is irrelevant.
If $\theta \in (0, \pi)$ then $\phi$ is unique.

The density matrix is of the form:

$$|\psi\rangle\langle\psi| = \begin{pmatrix} cos^2(\theta/2) & e^{-i\theta} sin(\theta/2) cos(\theta/2) \\ e^{i\theta} sin(\theta/2) cos(\theta/2) & sin^2(\theta/2) \end{pmatrix}$$

with the trigo formula we get:

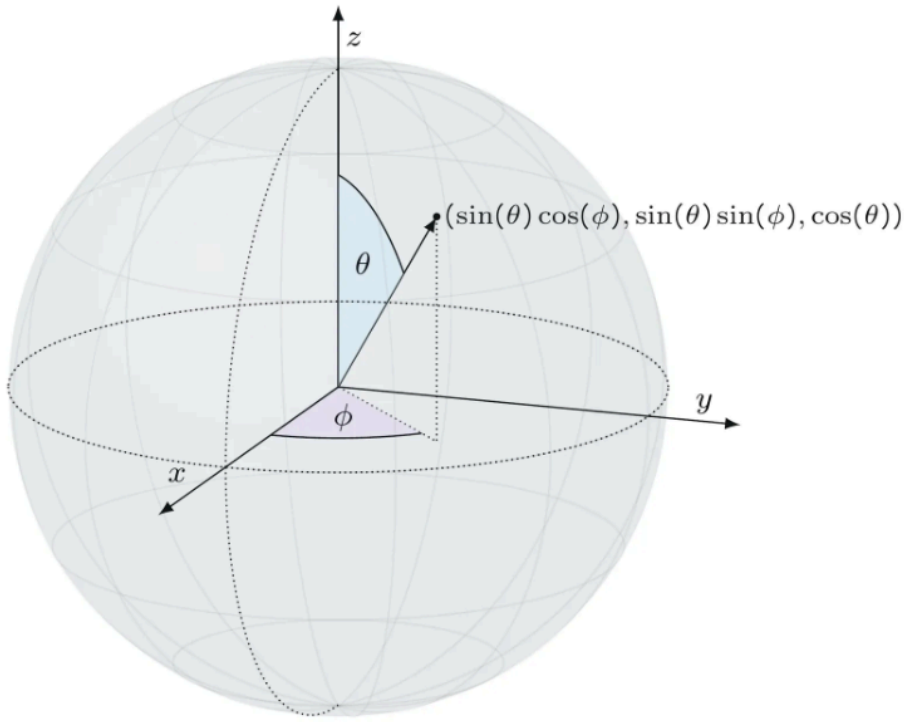
$$|\psi\rangle\langle\psi| = \begin{pmatrix} 1 + cos(\theta) & (cos(\theta) - isin(\theta)) * sin(\theta) \\ (cos(\theta) + isin(\theta)) * sin(\theta) & 1 - cos(\theta) \end{pmatrix}$$

In the Pauli basis with Pauli matrices:

$$\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

then the density matrix is a linear combination of Pauli matrices:

$$|\psi\rangle\langle\psi| = \frac{\mathbb{I} + sin(\theta)cos(\theta)\sigma_x + sin(\theta)sin(\theta)\sigma_y + cos(\theta)\sigma_z}{2}$$




This is the Bloch sphere representation of the $|\psi\rangle$ qubit. here are six important states:

$$|0\rangle\langle 0| = \frac{\mathbb{I} + \sigma_z}{2}$$

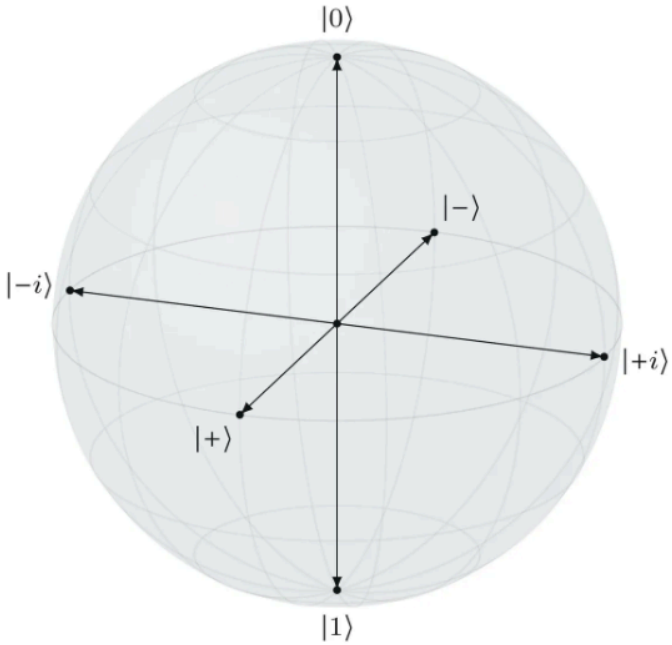
$$|1\rangle\langle 1| = \frac{\mathbb{I} - \sigma_z}{2}$$

$$|+\rangle\langle +| = \frac{\mathbb{I} + \sigma_x}{2}$$

$$|-\rangle\langle -| = \frac{\mathbb{I} + \sigma_x}{2}$$

$$|i\rangle\langle i| = \frac{\mathbb{I} + \sigma_y}{2}$$

$$|-i\rangle\langle -i| = \frac{\mathbb{I} - \sigma_y}{2}$$

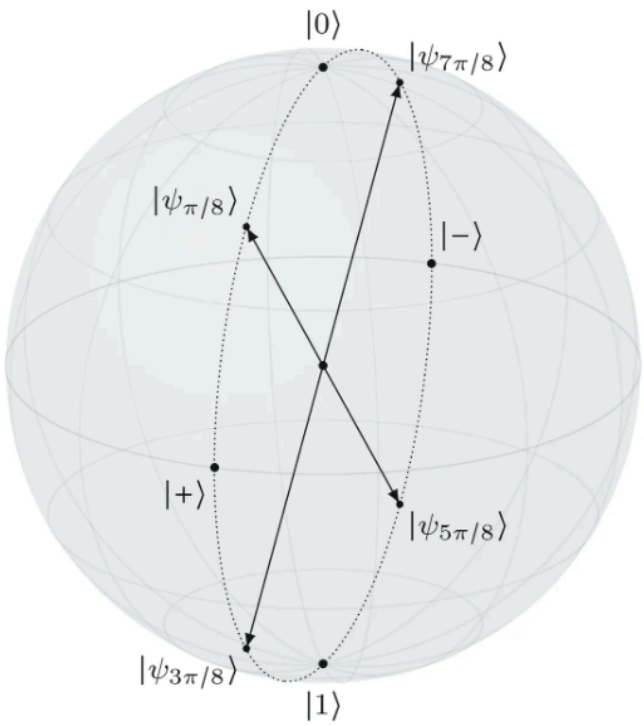



For a quantum state:

$$|\psi_a\rangle = cos(\alpha)|0\rangle + sin(\alpha)|1\rangle, \alpha \in [0, \pi)$$

The density matrix of such a state becomes :

$$|\psi_a\rangle\langle\psi_a| = \frac{\mathbb{I} + sin(2\alpha)\sigma_x + cos(2\alpha)\sigma_z}{2}$$




Points inside the Bloch sphere correspond to non-pure states. A convex combination of points in the Bloch sphere corresponds to a qubit matrix density combination. The point at the center of the Bloch sphere corresponds to $\mathbb{I}/2$.

# 4. Multiple systems and reduced states

Density matrices can represent multiple systems:

- Multiple systems are represented by a single matrix
- The rows and columns of the density matrix correspond to the cartesian product of the classical states of the individual systems.

Independence and correlation:

The tensor product of matrix densities represents the independence between systems.

Product states:

If $X$ is prepared in state $\rho$ and $Y$ is prepared independently in state $\sigma$, then state $(X, Y)$ is the tensor product $\rho \otimes \sigma$. States in this form are called product states.

Density matrices cannot be expressed as product states, which represent the correlation between states. Different types of correlations can be considered.

ex:

$$\frac{1}{2}|0\rangle\langle 0| \otimes |0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| \otimes |1\rangle\langle 1| = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{pmatrix}$$

States are not entangled, they are correlated. A convex combination of state products represents a classical correlation among quantum states.

$$\sum_{k=0}^{m-1} p_k \rho_k \otimes \sigma_k$$

States that can be written in this way are separable, while all others are intricate. State reduction for an e-bit. Alice and Bob share an e-bit, each with a qubit A and B respectively.

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

**Reduced state in general**

Now let's generalize the example just discussed to two arbitrary systems $A$ and $B$, not necessarily qubits in the state $|\phi^+\rangle$. We'll assume the classical state sets of $A$ and $B$ are $\Sigma$ and $\Gamma$, respectively. A density matrix $\rho$ representing a state of the combined system $(A, B)$ therefore has row and column indices corresponding to the Cartesian product $\Sigma \times \Gamma$.

Suppose that the state of $(A, B)$ is described by the quantum state vector $|\psi\rangle$, so

the density matrix describing this state is $\rho = |\psi\rangle\langle\psi|$. We'll obtain a density matrix description of the state of $A$ in isolation, which is conventionally denoted $\rho_A$.

The state vector $|\psi\rangle$ can be expressed in the form

$$|\psi\rangle = \sum_{b\in\Gamma} |\phi_b\rangle \otimes |b\rangle$$

for a uniquely determined collection of vectors $\{|\phi_b\rangle : b \in \Gamma\}$. In particular, these vectors can be determined through a simple formula.

$$|\phi_b\rangle = \big(\mathbb{I}_A \otimes \langle b|\big)|\psi\rangle$$

Reasoning similarly to the previous example of an e-bit, if we were to measure the system $B$ with a standard basis measurement, we would obtain each outcome $b \in \Gamma$ with probability $\||\phi_b\rangle\|^2$, in which case the state of $A$ becomes

$$\frac{|\phi_b\rangle}{\||\phi_b\rangle\|}.$$

As a density matrix, this state can be written as follows.

$$\left(\frac{|\phi_b\rangle}{\||\phi_b\rangle\|}\right)\left(\frac{|\phi_b\rangle}{\||\phi_b\rangle\|}\right)^\dagger = \frac{|\phi_b\rangle\langle\phi_b|}{\||\phi_b\rangle\|^2}$$

Averaging the different states according to the probabilities of the respective outcomes, we arrive at the density matrix

$$\rho_A = \sum_{b\in\Gamma} \||\phi_b\rangle\|^2 \frac{|\phi_b\rangle\langle\phi_b|}{\||\phi_b\rangle\|^2} = \sum_{b\in\Gamma} |\phi_b\rangle\langle\phi_b| = \sum_{b\in\Gamma} \big(\mathbb{I}_A \otimes \langle b|\big)|\psi\rangle\langle\psi|\big(\mathbb{I}_A \otimes |b\rangle\big)$$

**The partial trace**

The formula

$$\rho_{\mathsf{A}} = \sum_{b\in\Gamma} \big(\mathbb{I}_A \otimes \langle b|\big)|\psi\rangle\langle\psi|\big(\mathbb{I}_A \otimes |b\rangle\big)$$

leads us to the description of the reduced state of $\mathsf{A}$ for any density matrix $\rho$ of the pair $(\mathsf{A}, \mathsf{B})$, not just a pure state.

$$\rho_A = \sum_{b\in\Gamma} \big(\mathbb{I}_A \otimes \langle b|\big)\rho\big(\mathbb{I}_A \otimes |b\rangle\big)$$

The operation being performed on $\rho$ to obtain $\rho_A$ in this equation is known as the *partial trace*, and to be more precise we say that the partial trace is performed on $B$, or that $B$ is *traced out*. This operation is denoted $Tr_B$, so we can write

$$Tr_B(\rho) = \sum_{b\in\Gamma} \big(\mathbb{I}_A \otimes \langle b|\big)\rho\big(\mathbb{I}_{\mathsf{A}} \otimes |b\rangle\big).$$

We can also define the partial trace on $A$, so it's the system $A$ that gets traced out rather than $B$, like this.

$$Tr_A(\rho) = \sum_{a \in \Sigma} \big(\langle a| \otimes \mathbb{I}_B\big)\rho\big(|a\rangle \otimes \mathbb{I}_B\big)$$

**Alternative description of the partial trace**

An alternative way to describe the partial trace mappings $Tr_A$ and $Tr_B$ is that they are the *unique* linear mappings that satisfy the formulas

$$Tr_A(M \otimes N) = Tr(M)N$$
$$Tr_B(M \otimes N) = Tr(N)M.$$

# 5. Implementation on qiskit

In [1]:
```
# Required imports for code cells

from qiskit.quantum_info import Statevector, Operator, DensityMatrix
from qiskit.visualization import array_to_latex
import numpy as np
from numpy import linalg

print('Imports loaded.')
```

```
Imports loaded.
```

In [2]:
```
# Randomly generate a positive semidefinite matrix

n = 3
M = np.random.randint(-9, 10, size=(n, n)) + 1j * np.random.randint(-9, 1
P = M.conj().T @ M

display(array_to_latex(P))
```

$$\begin{bmatrix} 121 & -72 - 8i & 5 - 11i \\ -72 + 8i & 299 & -157 - 64i \\ 5 + 11i & -157 + 64i & 117 \end{bmatrix}$$

In [3]:
```
# Compute the eigenvalues (in decreasing order)

display(array_to_latex(np.sort(linalg.eigvals(P))[::-1]))
```

$$\begin{bmatrix} 413.7447410866 & 121.7456791518 & 1.5095797616 \end{bmatrix}$$

In [4]:
```
# Randomly generate a density matrix

n = 3
M = np.random.randint(-9, 10, size=(n, n)) + 1j * np.random.randint(-9, 1
```

```
P = M.conj().T @ M
rho = P/np.trace(P)

display(array_to_latex(rho))
```

$$\begin{bmatrix} 0.4319775596 & \frac{2}{31} + \frac{i}{23} & 0.255259467 \\ \frac{2}{31} - \frac{i}{23} & 0.3071528752 & 0.1654978962 - 0.0014025245i \\ 0.255259467 & 0.1654978962 + 0.0014025245i & \frac{6}{23} \end{bmatrix}$$

In [5]:
```
# Compute the eigenvalues (in decreasing order)

display(array_to_latex(np.sort(linalg.eigvals(rho))[::-1]))
```

$$\begin{bmatrix} 0.6813602138 & \frac{40 \cdot 2^{\frac{81}{205}} \cdot 3^{\frac{27}{41}} \cdot 5^{\frac{159}{205}} \cdot 7^{\frac{178}{205}}}{7203} & 0.0344474666 \end{bmatrix}$$

In [6]:
```
# Initialize and display a density matrix
# using the DensityMatrix package

rho = DensityMatrix([[3/4, 1j/8],[-1j/8,1/4]])
display(rho.draw('latex'))
```

$$\begin{bmatrix} \frac{3}{4} & \frac{i}{8} \\ -\frac{i}{8} & \frac{1}{4} \end{bmatrix}$$

In [7]:
```
# Define and display examples of density matrices
# using the Statevector package

ket_0 = Statevector.from_label('0')
ket_1 = Statevector.from_label('1')
ket_plus = Statevector.from_label('+')
ket_minus = Statevector.from_label('-')
ket_iplus = Statevector.from_label('r')
ket_iminus = Statevector.from_label('l')
v = Statevector([(1 + 2j) / 3, -2 / 3])

display(array_to_latex(ket_0.to_operator()))
display(array_to_latex(ket_1.to_operator()))
display(array_to_latex(ket_plus.to_operator()))
display(array_to_latex(ket_minus.to_operator()))
display(array_to_latex(ket_iplus.to_operator()))
display(array_to_latex(ket_iminus.to_operator()))
display(array_to_latex(v.to_operator()))
```

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & -\frac{i}{2} \\ \frac{i}{2} & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{5}{9} & -\frac{2}{9} - \frac{4i}{9} \\ -\frac{2}{9} + \frac{4i}{9} & \frac{4}{9} \end{bmatrix}$$

In [8]:

```
# Define and display examples of density matrices
# using the DensityMatrix package

rho_0 = DensityMatrix.from_label('0')
rho_1 = DensityMatrix.from_label('1')
rho_plus = DensityMatrix.from_label('+')
rho_minus = DensityMatrix.from_label('-')
rho_iplus = DensityMatrix.from_label('r')
rho_iminus = DensityMatrix.from_label('l')

display(rho_0.draw('latex'))
display(rho_1.draw('latex'))
display(rho_plus.draw('latex'))
display(rho_minus.draw('latex'))
display(rho_iplus.draw('latex'))
display(rho_iminus.draw('latex'))
```

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

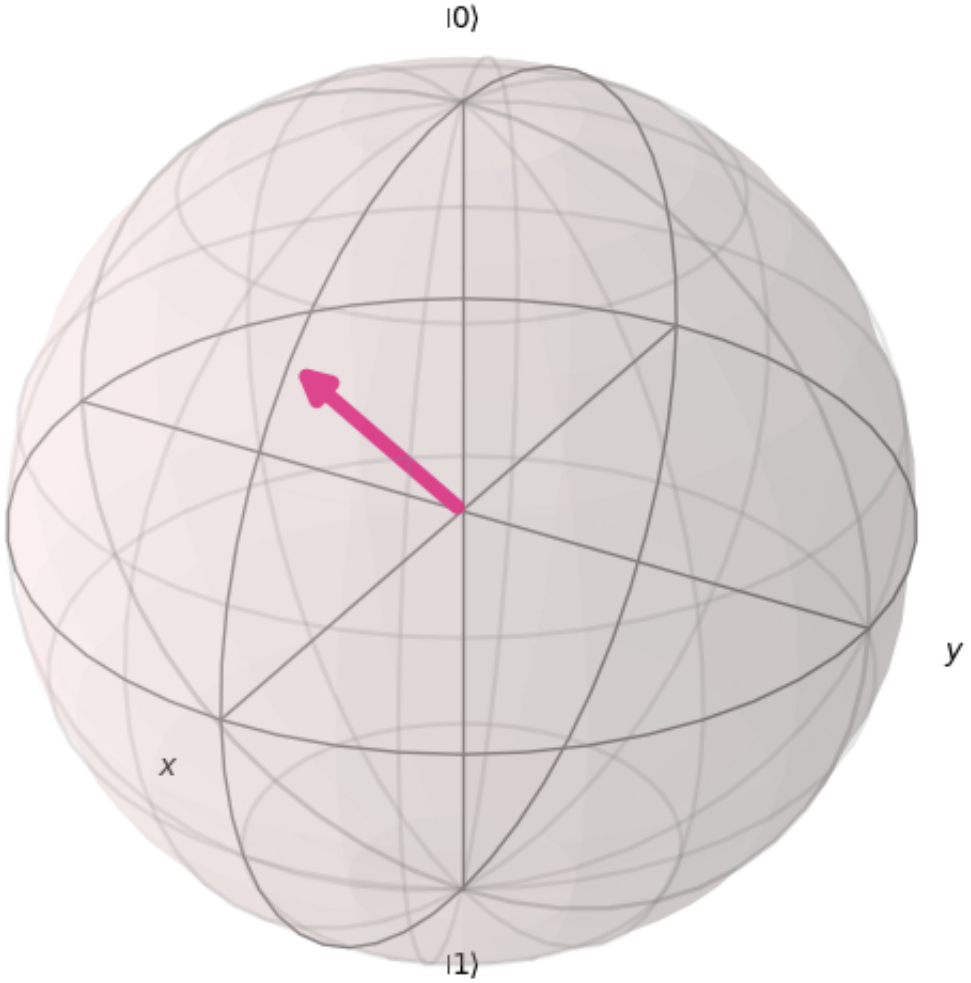
$$\begin{bmatrix} \frac{1}{2} & -\frac{i}{2} \\ \frac{i}{2} & \frac{1}{2} \end{bmatrix}$$

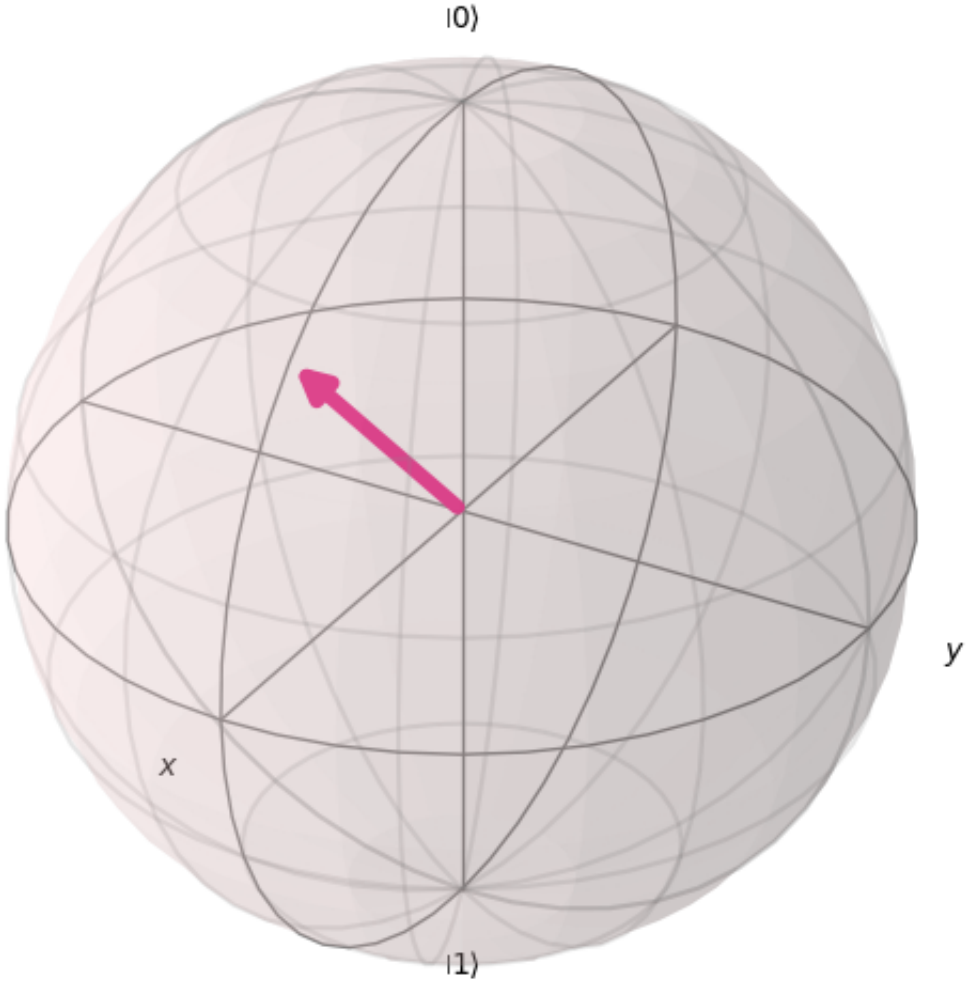
$$\begin{bmatrix} \frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{bmatrix}$$

In [9]:
```
# Required imports for code cell

from qiskit.quantum_info import DensityMatrix
from qiskit.visualization import plot_bloch_multivector, plot_bloch_vecto
from numpy import sin, cos, pi

print('loaded')
```

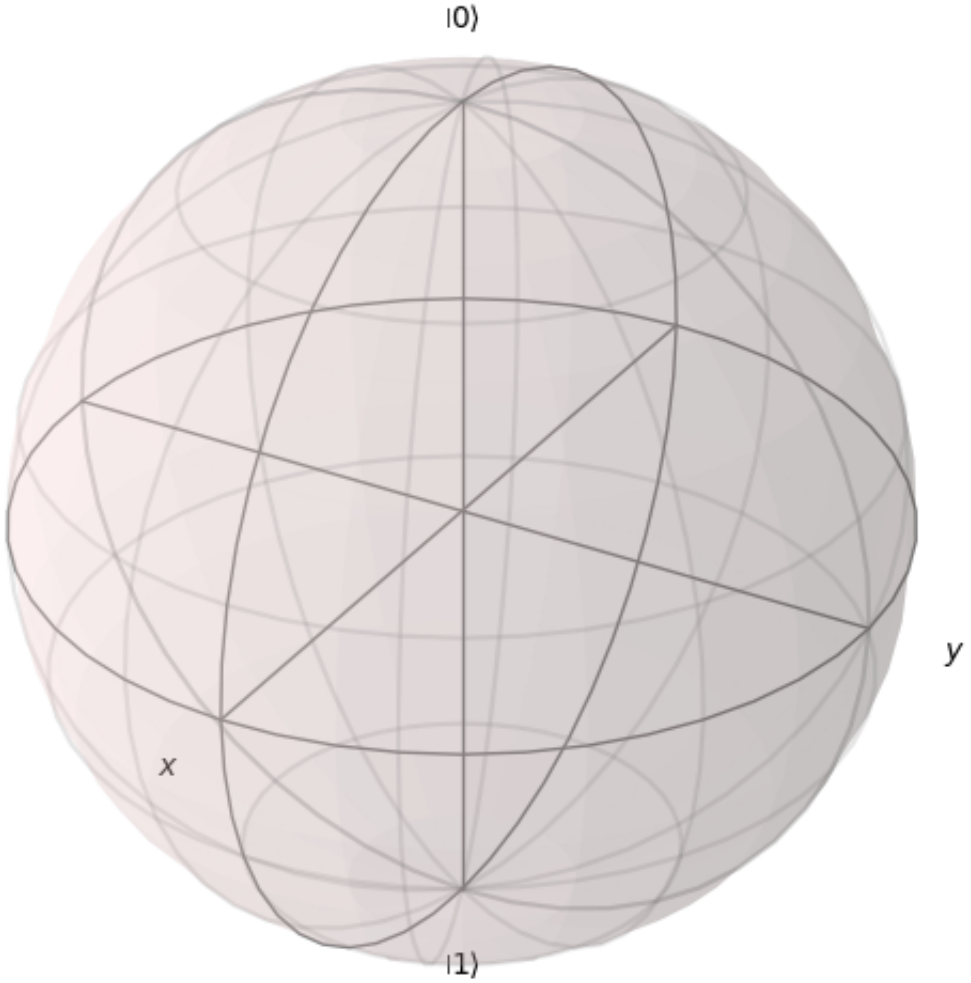
```
loaded
```

In [10]:
```
display(plot_bloch_vector([sin(pi/4),0,cos(pi/4)]))
display(plot_bloch_vector([1, pi/4, 0], coord_type='spherical'))
```

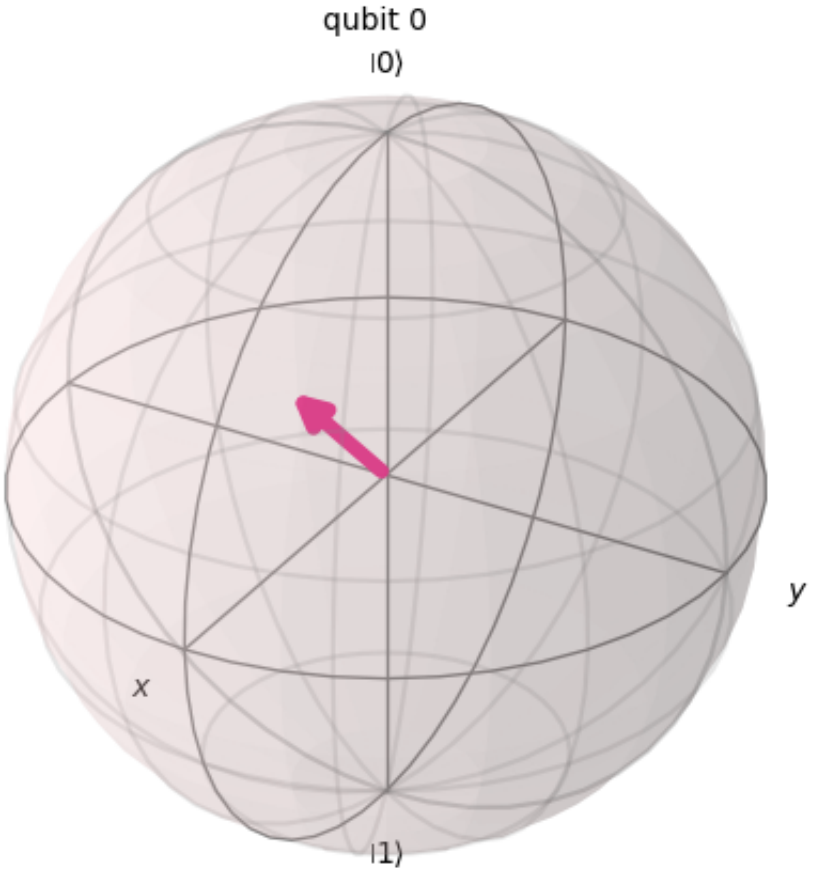

|0⟩
y
x
|1⟩



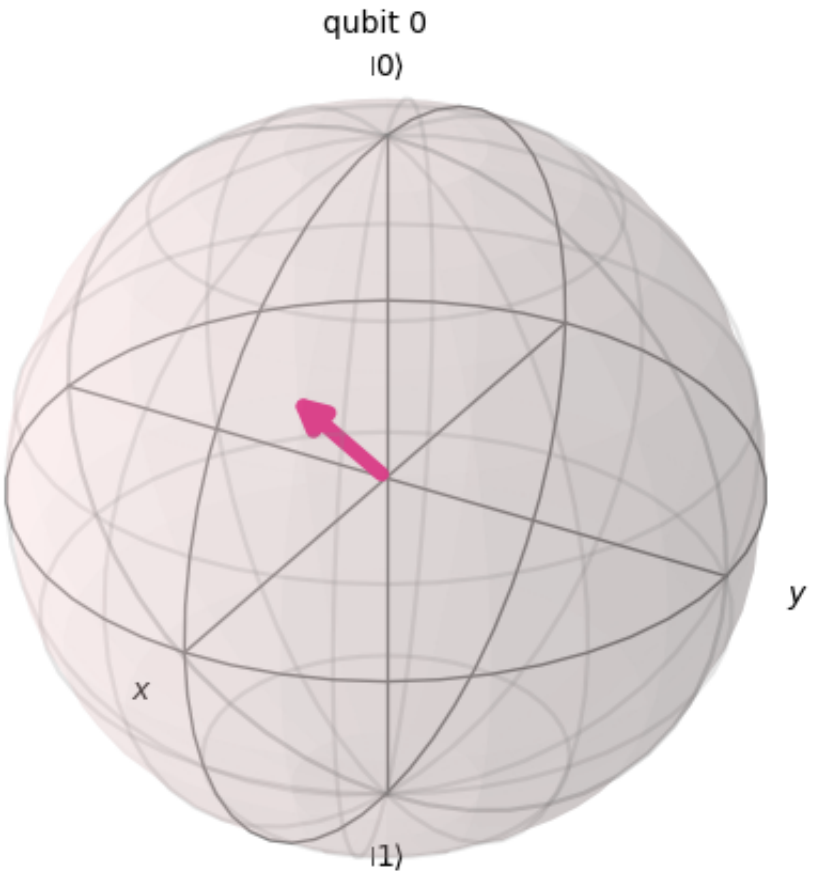

In [11]:
```
# Plot the center point of the Bloch ball
# using plot_bloch_vector

display(plot_bloch_vector([0,0,0]))
```




In [12]:
```
# Plot a Bloch ball point using plot_bloch_multivector

rho = DensityMatrix([[3/4, 1/4],[1/4,1/4]])
display(plot_bloch_multivector(rho))
```




In [13]:
```
# Plot a Bloch ball point using the
# DensityMatrix draw method

display(rho.draw('bloch'))
```




## Reduced states in qiskit

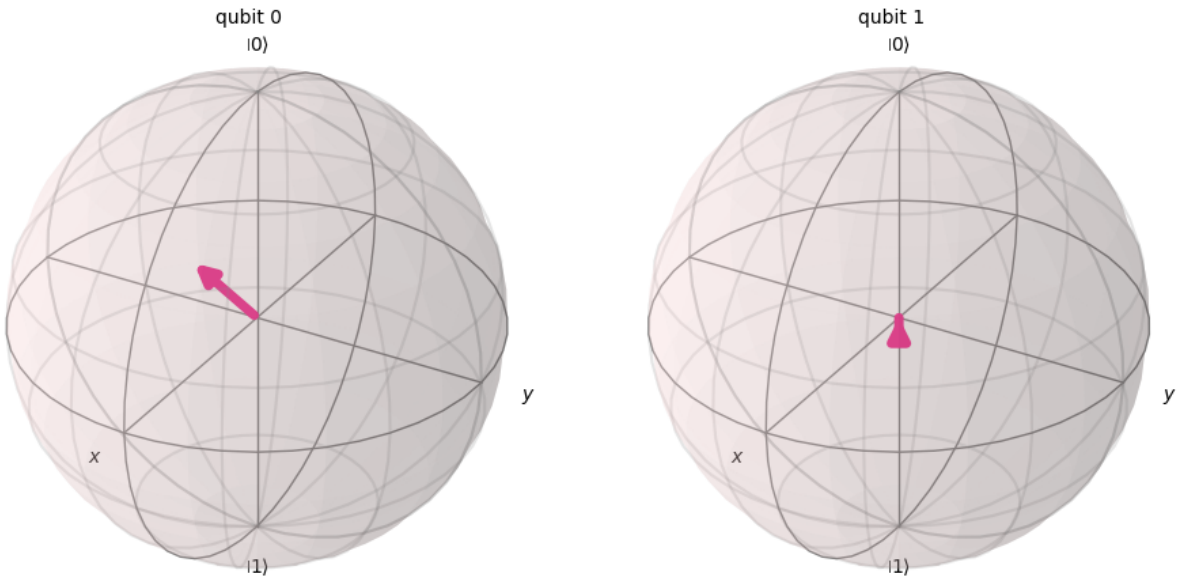
In [14]:
```
from qiskit.quantum_info import DensityMatrix, partial_trace
from qiskit.visualization import plot_bloch_multivector
print('loaded')
```

```
loaded
```

In [15]:
```
rho_0 = DensityMatrix.from_label('0')
rho_1 = DensityMatrix.from_label('1')
rho_plus = DensityMatrix.from_label('+')
rho = 1/2*rho_0.tensor(rho_0) + 1/2*rho_1.tensor(rho_plus)

display(rho.draw('latex'))
display(partial_trace(rho, [0]).draw(output='latex'))
display(partial_trace(rho, [1]).draw(output='latex'))
```

$$\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

In [16]:
```
display(plot_bloch_multivector(rho))
```